

Clean Geodesic Return and Exact Laplace Revival on Round Spheres

Route-A source-local certificate HCS-C313

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Abstract

For every round sphere S_R^d , $d \geq 2$, we solve the unit tangent flow, identify its oriented-Grassmann orbit quotient, and prove that every orbit has least period $2\pi R$ with maximally clean identity return. We then derive the full Laplace spectrum and heat trace and prove exact scalar and identity revivals for the natural completed-square half-wave.

1 Classical flow, orbit quotient, and clean return

Embed S_R^d in $\mathbb{R}^{d+1}$ and write

$$T^1 S_R^d = \{(x, v) : |x| = R, |v| = 1, x \cdot v = 0\}.$$

Theorem 1 (Complete round geodesic flow). *The unit-speed geodesic flow is*

$$\Phi_t(x, v) = (\cos(t/R)x + R \sin(t/R)v, -R^{-1} \sin(t/R)x + \cos(t/R)v). \quad (1)$$

Every orbit has least period $L = 2\pi R$. More precisely,

$$\text{Fix}(\Phi_t) = \begin{cases} T^1 S_R^d, & t \in L\mathbb{Z}, \\ \emptyset, & t \notin L\mathbb{Z}. \end{cases} \quad (2)$$

At every return $D\Phi_t = I$, so (2) is clean with the largest possible fixed manifold. The orbit space is the oriented Grassmannian $G_2^+(\mathbb{R}^{d+1})$; the unit tangent bundle is its principal circle bundle. The flow preserves Liouville measure and is reversed by $\iota(x, v) = (x, -v)$.

Proof. The two displayed vectors keep norms $R, 1$, remain orthogonal, and the first derivative is the second; its second derivative is $-R^{-2}x(t)$, the round geodesic equation. On the oriented plane spanned by the orthonormal frame $(x/R, v)$, (1) is rotation through t/R . It fixes that frame exactly at multiples of 2π , proving least period and (2); the ambient rotation is then the identity, so is its differential. Quotienting oriented frames by this circle rotation gives G_2^+ , of dimension $2d - 2$. Hamiltonian geodesic flow preserves Liouville measure, and direct substitution gives $\iota\Phi_t\iota = \Phi_{-t}$. $\square$

2 Laplace spectrum, heat trace, and exact revival

Let $-\Delta \geq 0$ be the Laplace–Beltrami operator. Restrictions of homogeneous harmonic polynomials of degree ℓ form $\mathcal{H}_\ell$.

Theorem 2 (Complete source quantization). *For every $\ell \geq 0$,*

$$-\Delta|_{\mathcal{H}_\ell} = \frac{\ell(\ell + d - 1)}{R^2}, \quad m_{d,\ell} = \dim \mathcal{H}_\ell = \frac{(2\ell + d - 1)(\ell + d - 2)!}{\ell!(d - 1)!}. \quad (3)$$

The spaces $\mathcal{H}_\ell$ form a complete orthogonal decomposition of $L^2(S_R^d)$. Consequently, for $s > 0$,

$$\text{Tr}(e^{s\Delta}) = \sum_{\ell=0}^{\infty} m_{d,\ell} e^{-s\ell(\ell+d-1)/R^2} < \infty. \quad (4)$$

Define by spectral calculus

$$Q_d = \sqrt{-\Delta + \frac{(d-1)^2}{4R^2}}. \quad (5)$$

Then

$$e^{-i2\pi R Q_d} = (-1)^{d-1} I, \quad e^{-i4\pi R Q_d} = I. \quad (6)$$

Proof. The Euclidean polar decomposition of homogeneous polynomials gives the eigenvalue in (3). Subtracting the degree- $\ell - 2$ polynomials divisible by $|x|^2$ gives $m_{d,\ell} = \binom{\ell+d}{d} - \binom{\ell+d-2}{d}$, equal to the displayed factorial. Polynomial density gives completeness. Since $m_{d,\ell} = O(\ell^{d-1})$ and the eigenvalue is quadratic, (4) converges. Finally,

$$\ell(\ell + d - 1) + (d - 1)^2/4 = (\ell + (d - 1)/2)^2,$$

so Q_d has frequency $(\ell + (d - 1)/2)/R$. Substitution proves both phases in (6). $\square$

The shift in (5) is a source functional-calculus completion of the square. No exact revival is claimed here for the unshifted $\sqrt{-\Delta}$, and no target spectrum is attached to either operator.

Source lineage

References

- [1] Y. Colin de Verdière, “Spectrum of the Laplace operator and periodic geodesics: thirty years after,” *Ann. Inst. Fourier* 57 (2007), 2429–2463. DOI: [10.5802/aif.2339](https://doi.org/10.5802/aif.2339).